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Paper 1: PMxAgent: An Agentic Platform for Pharmacometrics

Bibliographic Information

- **Title:** PMxAgent: An Agentic Platform for Pharmacometrics
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Paper Type: METHODOLOGY

Quick Take

PMxAgent is an open-source agentic platform that exposes validated R-based pharmacometric tools (NCA, ER, PK simulation, data standardization, model library) as MCP-compatible tools for AI agents. In a benchmark across 182 drugs and 1820 subjects, PMxAgent achieved 98.3% NCA accuracy, matching or exceeding frontier AI agents, while producing deterministic, reproducible results. The platform separates AI orchestration from deterministic scientific execution, addressing reproducibility concerns in AI-driven pharmacometrics.

Executive Summary

PMxAgent provides a containerized architecture (Docker) with an R API server (Plumber) and a Python MCP server (FastMCP) that auto-generates agent-callable tools from OpenAPI specifications. Five pharmacometric tools were developed and validated with 96 automated tests. A proof-of-concept case study demonstrated autonomous orchestration of PK simulation, data standardization, NCA, and ER analysis. The NCA benchmark showed PMxAgent's accuracy (98.3%) comparable to Claude Opus 4.8 (98.2%) and superior to GPT 5.5 variants, with bit-for-bit reproducibility across runs. The platform's design ensures deterministic execution, traceability, and transparency, making it suitable for GxP-regulated workflows.

Scientific Context & Motivation

Pharmacometric analyses rely on diverse software and manual workflows, leading to reproducibility and standardization challenges. AI agents offer potential for automation but are nondeterministic, complicating validation. There is no standard framework for integrating validated pharmacometric tools into agentic workflows. PMxAgent addresses this gap by providing a protocol-based (MCP) platform that exposes deterministic R tools to AI agents, separating orchestration from execution.

✂ Methodological Snapshot

PMxAgent is a containerized platform with two services: an R API server (Plumber) exposing pharmacometric functions as REST endpoints with OpenAPI specs, and a Python MCP server

(FastMCP) that auto-generates agent-callable tools from these specs. Five tools were developed: NCA (PKNCA), ER analysis (linear, Emax, Imax, logistic with AIC selection), PK simulation (mrgsolve), data standardization (CDISC ADaM ADPC), and model library (nlmixr2lib). The platform uses Docker Compose for orchestration, with bind-mounted directories for file-based data exchange. AI agents (e.g., Cursor, Claude Code) discover and invoke tools via MCP, while all scientific computation runs deterministically in R.

⌘ Statistical Framework

The platform relies on established pharmacometric methods: NCA uses PKNCA's trapezoidal rule (linear-up log-down) with best adjusted-R2 terminal slope selection and configurable BLQ handling; ER analysis fits multiple models (linear, Emax, Imax, logistic) and selects by AIC; PK simulation uses mrgsolve with one- or two-compartment IV bolus models and between-subject variability; data standardization follows CDISC ADaM ADPC conventions. The statistical framework assumes standard PK/PD modeling assumptions (e.g., log-linear terminal phase, independent subjects). The platform's determinism is achieved by executing the same R code for each tool call, independent of the AI agent's stochasticity.

Estimator Behavior

The NCA tool's estimators (AUC, Cmax, Tmax, half-life, CL, Vz) are computed via PKNCA, which is validated against PKanalix. In the benchmark, PMxAgent achieved 98.3% accuracy ($|\text{RE}| < 1\%$) across seven parameters, with Cmax and Tmax 100% accurate. Errors were minimal and confined to λ_z -dependent parameters due to terminal-point selection. The platform is deterministic: 10 repeated runs produced bit-for-bit identical results. In contrast, GPT 5.5 pro showed bimodal accuracy (92.7%-95.6%) due to stochastic terminal-point selection.

Validation Design

Validation included 96 automated tests: 63 Python integration tests (endpoint, validation, library) and 33 R unit tests (NCA ground truth, PK model analytical solutions, library). The NCA benchmark used 182 drugs from nlmixr2lib, 1820 simulated subjects, with PKanalix as reference. Accuracy was defined as $|\text{RE}| < 1\%$ for each parameter, with subjects having adjusted $R^2 < 0.80$ excluded from λ_z -dependent parameters. Reproducibility was assessed by 10 repeated runs for PMxAgent and GPT 5.5 pro.

Applicability Boundaries

PMxAgent works well for NCA on clean, structured data with known dosing and sampling times. It is designed for single-user local deployments; enterprise use requires additional infrastructure. The platform's tools are limited to the five implemented functions; more complex analyses (e.g., model building, covariate analysis) are not yet available. The benchmark used simulated data without data cleaning tasks, so performance on messy real-world data is untested. The file-based data exchange is efficient for large datasets but requires shared storage.

Comparison to Alternatives

Compared to frontier AI agents (GPT 5.5, Claude Opus 4.8, Claude Sonnet 4.6) that generate code for NCA, PMxAgent provides deterministic execution and higher or comparable accuracy (98.3% vs 91.6%-98.2%). It also offers traceability and reproducibility that code-generating agents lack. However, code-generating agents are more flexible for exploratory analyses. PMxAgent’s tool-based approach is less flexible but more reliable for regulated workflows.

Implementation Guidance

Deployment requires Docker Desktop; run ‘docker compose up –build’. The R API is exposed at localhost:5762, MCP server at localhost:8000. Tools are auto-discovered by MCP-compatible clients (e.g., Cursor, Claude Code). Users can add new R tools by creating Plumber endpoints and updating the OpenAPI spec. The platform uses OAuth 2.0 for authentication (auto-approved locally). For production, additional security and audit features are needed. Computational cost is low for the included tools; NCA of 1820 subjects runs in seconds.

☒ Key Findings

- 1) PMxAgent enables AI agents to autonomously orchestrate multistep pharmacometric workflows (PK simulation, data standardization, NCA, ER) with human oversight.
 - 2) In a large-scale benchmark, PMxAgent achieved 98.3% overall NCA accuracy ($|RE| < 1\%$) across 1820 subjects, matching Claude Opus 4.8 (98.2%) and exceeding GPT 5.5 pro (95.6%), Claude Sonnet 4.6 (94.5%), and GPT 5.5 thinking (91.6%).
 - 3) PMxAgent produced bit-for-bit identical results across 10 runs, while GPT 5.5 pro varied bimodally (92.7%-95.6%).
 - 4) Errors in frontier agents were primarily due to terminal slope selection and BLQ handling, affecting λ_z -dependent parameters.
 - 5) The platform’s file-based data exchange avoids context window limitations and transcription errors.
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Strengths & Limitations

Strengths

- Deterministic execution of validated R code ensures reproducibility and traceability.
- Modular architecture (R API + MCP server) allows easy addition of new tools and integration with any MCP-compatible agent.
- Comprehensive automated testing (96 tests) covering integration and unit levels.
- Large-scale benchmark (182 drugs, 1820 subjects) against validated reference (PKanalix) provides robust accuracy assessment.
- File-based data exchange reduces token usage and transcription errors.
- Open-source and containerized for portability.

Limitations (Acknowledged by Authors)

- Benchmark used simulated, clean data; no data cleaning tasks included.

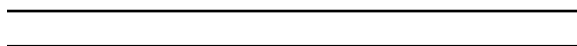
- Structured prompt was necessary for fair comparison; BLQ handling instructions inadvertently omitted.
- Enterprise/regulated deployment requires additional features (RBAC, audit trails, GxP validation).
- File-based data exchange suited for single-user local deployments; multiuser needs cloud storage.
- Agent-level errors (tool discovery, parameter handling, summarization) and human oversight still required.

Limitations (Expert Review)

- The benchmark only assessed analytical accuracy on NCA-ready data; real-world data wrangling and unit handling not evaluated.
- The platform’s determinism relies on the underlying R code; any changes to R packages or dependencies could affect reproducibility.
- The case study used a small dataset (60 subjects); scalability to larger datasets and more complex workflows remains untested.
- The ER analysis used a simple binary response; more complex exposure-response models (e.g., time-varying) not covered.
- The platform’s tool scope is limited to the five examples; broader pharmacometric tasks (e.g., model building, covariate analysis) not yet implemented.

Generalizability

The architecture is generalizable to any R-based pharmacometric tool and can be extended to other languages via HTTP/OpenAPI. However, the benchmark results are specific to NCA on simulated data; generalizability to other analyses and real-world data requires further validation.



Figures & Tables

- **Figure 1:** System architecture of PMxAgent showing the two-container setup (R API server and Python MCP server) and data flow.
 - *Significance:* Illustrates the core architectural design that enables deterministic tool execution and agent integration.
- **Figure 2:** Cursor agent response showing decomposition of the case study prompt into four sequential to-do steps and initiation of the PK simulation tool call.
 - *Significance:* Demonstrates the agent’s ability to plan and execute multistep workflows with human approval.
- **Figure 3:** Simulated concentration-time profiles from the two-compartment IV bolus PK model for 60 subjects across three dose groups.
 - *Significance:* Shows the output of the PK simulation tool and the between-subject variability.
- **Figure 4:** Exposure-response analysis results showing individual binary outcomes, model fits, and AIC-based model selection.
 - *Significance:* Demonstrates the ER tool’s automatic model selection and parameter estimation.

- **Figure 5:** Structured case study results summary generated by the agent, documenting outputs from each tool in the workflow.
 - *Significance:* Highlights the platform’s ability to produce comprehensive, publication-ready reports.
 - **Figure 6:** Overall NCA accuracy across AI agents (PMxAgent, GPT 5.5 pro, GPT 5.5 thinking, Claude Opus 4.8, Claude Sonnet 4.6).
 - *Significance:* Key benchmark result showing PMxAgent’s accuracy (98.3%) relative to frontier agents.
 - **Figure 7:** Per-subject error distributions by parameter and agent, excluding Cmax and Tmax which were 100% accurate.
 - *Significance:* Identifies terminal slope selection and BLQ handling as primary sources of error in other agents.
 - **Table 1:** Description of the five pharmacometric tools (NCA, ER, PK, DATA, LIBRARY) including their capabilities and inputs.
 - *Significance:* Provides a detailed reference for the tool functionalities and usage.
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Code & Reproducibility Assessment

All code, Dockerfiles, and example data are available at <https://github.com/peterbloomingle/PMxAgent>. The README provides instructions for local reproduction. The platform is designed for portability across OS and hardware.

Supplementary Materials

Supplementary materials include Figures S1 and S2 (reproducibility analysis for PMxAgent and GPT 5.5 pro) and Appendix S1 (benchmark prompt).

Future Directions

Future work should extend benchmarking to data cleaning, unit handling, and prompt sensitivity. Enterprise features (RBAC, audit logging, cloud storage) are needed for regulated deployment. Expanding the tool library to include model building, covariate analysis, and simulation-based diagnostics would increase utility. Standardization of tool design and nomenclature across the pharmacometrics community is also needed.

Expert Commentary

This paper addresses a critical gap in AI-driven pharmacometrics by separating deterministic execution from AI orchestration. The benchmark design is rigorous, using a validated reference and large simulated dataset. The finding that a lower-cost model (Claude Sonnet 4.6) augmented with tools outperforms its direct computation is compelling evidence for tool augmentation. The platform’s architecture is forward-looking, leveraging MCP and OpenAPI standards. However, real-world adoption will require addressing enterprise security and scalability, and the

benchmark’s focus on clean simulated data limits immediate generalization. Nonetheless, PMx-Agent is a significant step toward reproducible AI-augmented pharmacometrics.

Bottom Line

PMxAgent offers a practical, open-source framework for integrating validated pharmacometric tools into AI-agent workflows, ensuring deterministic, reproducible, and traceable analyses. It demonstrates that tool-augmented agents can match or exceed frontier models in accuracy while providing the reproducibility needed for regulatory applications. Pharmacometricians can adopt this platform to accelerate workflows while maintaining scientific oversight.

☒ Figures

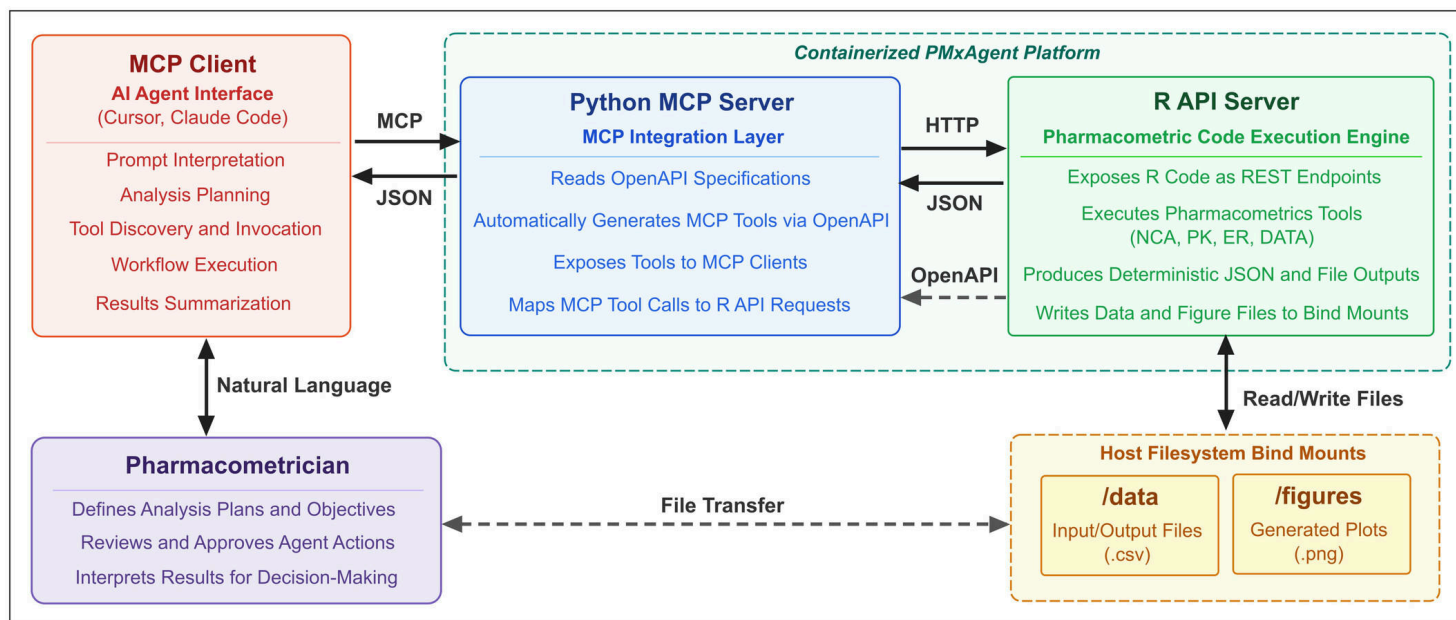


Figure 1: System architecture of PMxAgent. PMxAgent is deployed as a containerized platform with two core services: A Python MCP server and an R API server. The Python MCP

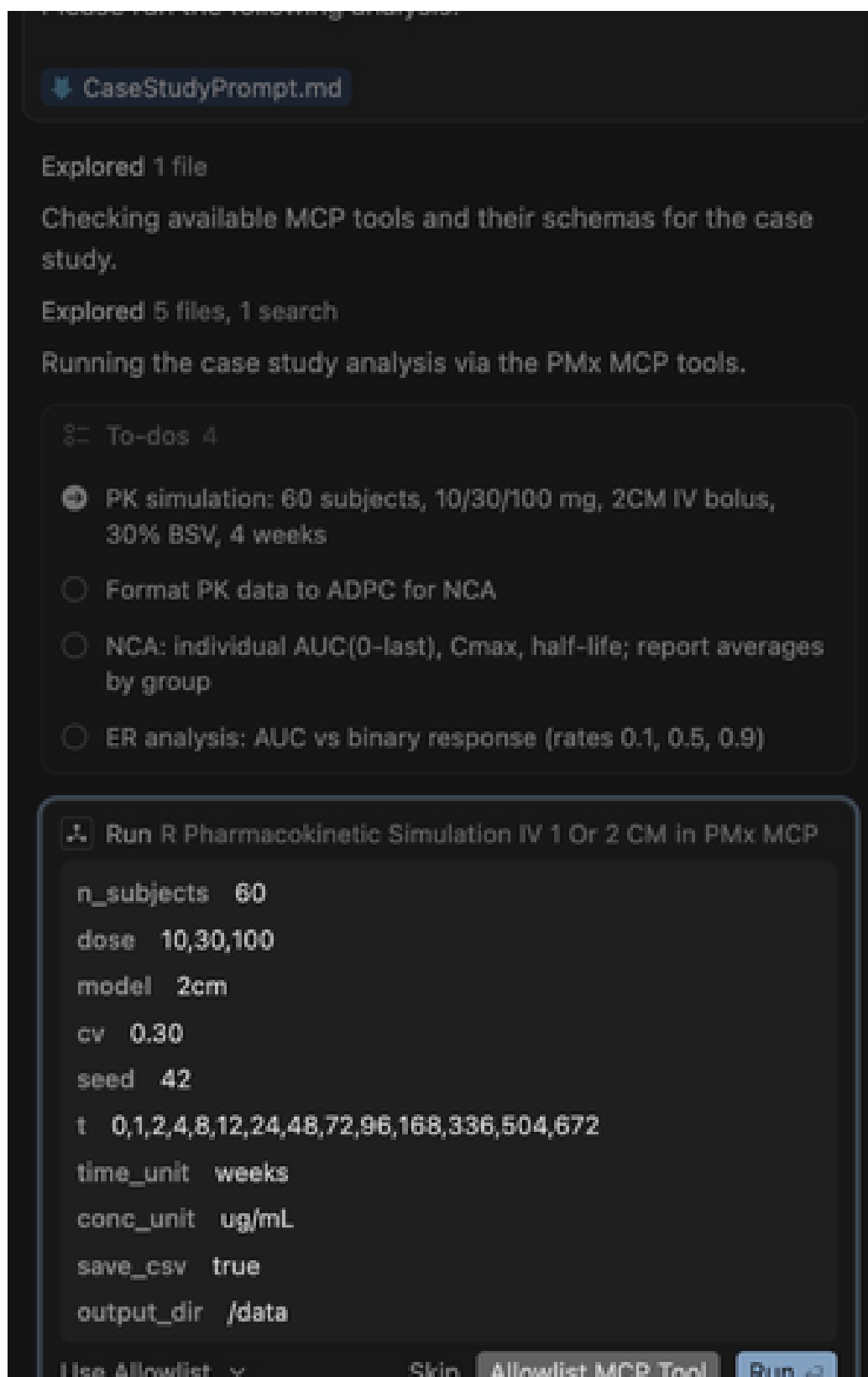


Figure 2: Cursor (Composer 1.5) agent response showing decomposition of the case study prompt into four sequential to-do steps and the initiation of Step 1: A PK simulation

Population PK Simulation (two-compartment, Bolus)

Mean $\pm$ SD, n = 20 per group

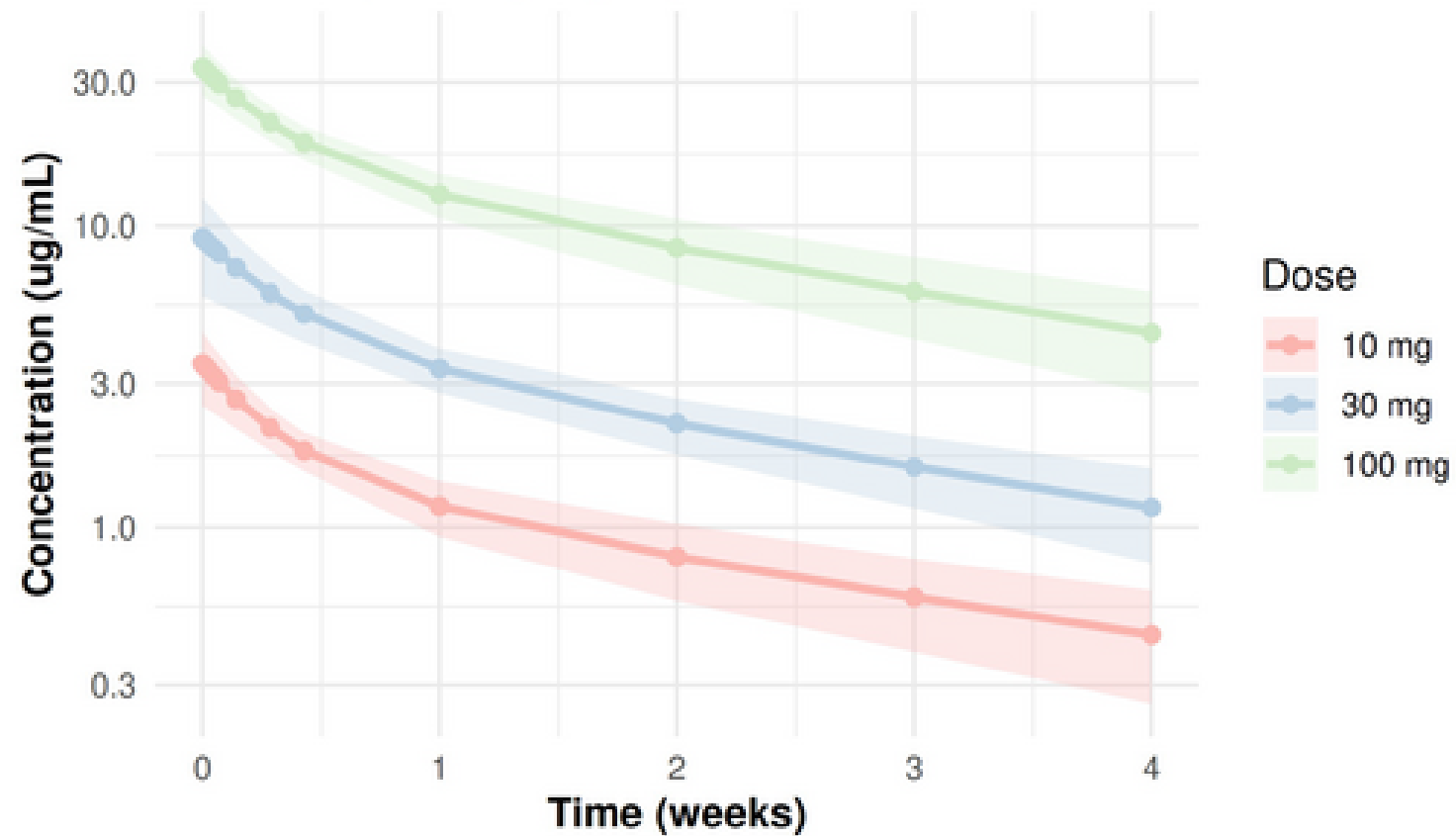


Figure 3: Simulated concentration-time profiles from the two-compartment IV bolus PK model for 60 subjects (20 per dose group) at doses of 10, 30, and 100 mg. Lines repres

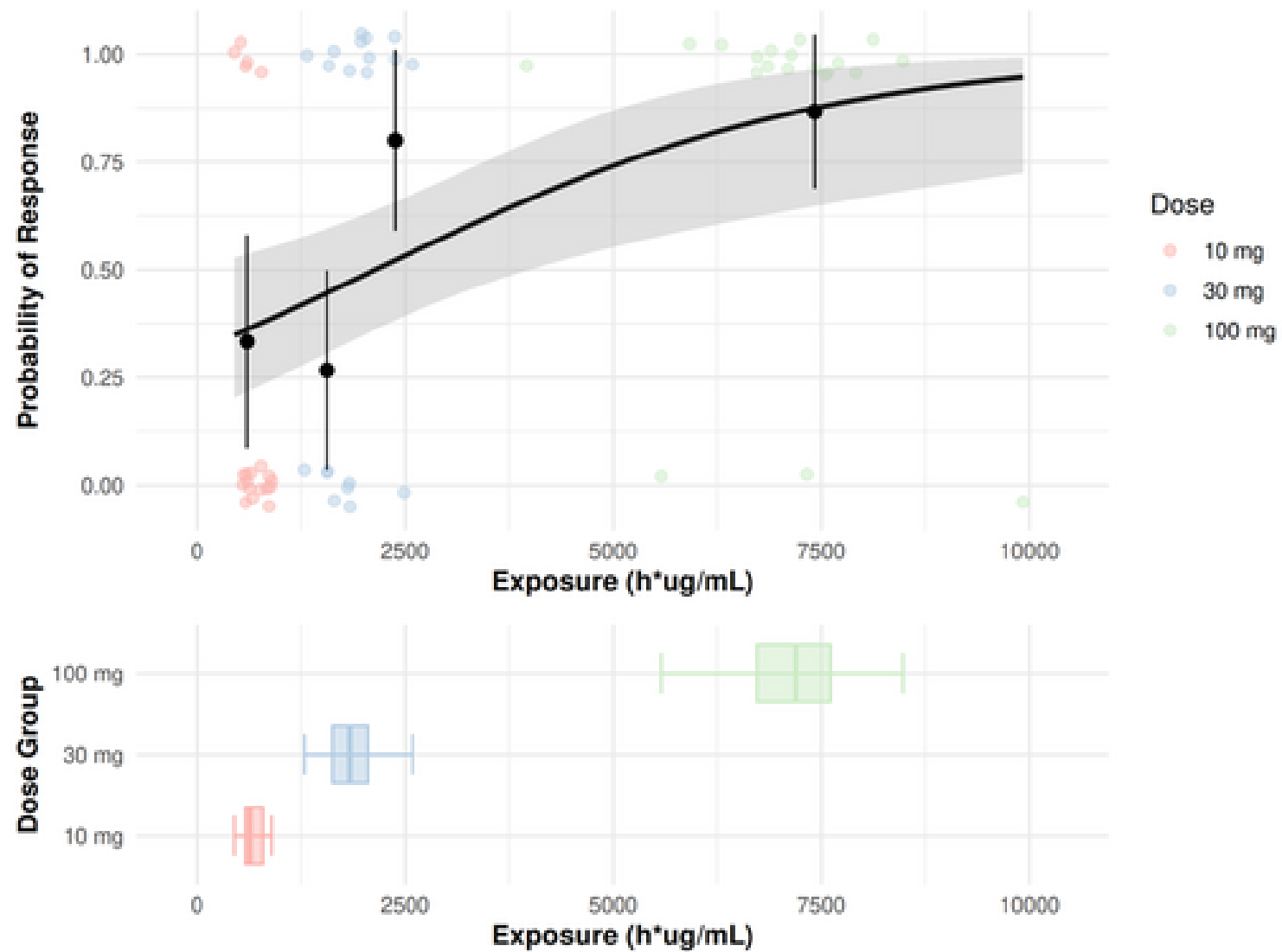


Figure 4: Exposure-response analysis results from the case study. Upper panel: Individual binary response outcomes (colored points) and dose-group mean $\pm$ SD (black points)

PMxAgent Case Study Results

Workflow: PK simulation → DATA standardization → NCA → ER analysis (seed 42 throughout).

(1) PK Simulation

- Design: 60 subjects (20/dose), doses 10 / 30 / 100 mg, IV bolus 2-compartment model, 30% BSV, 4-week window (0–672 h)
- Model: Betts 2018 typical values (CL 0.15 mL/h/kg, V1 46.31 mL/kg, V2 31.47 mL/kg, Q 0.27 mL/h/kg)
- Plot: `/figures/PK_20260617_032912.png`

(2) NCA — Group Summaries (IV bolus, AUC 0–last)

Dose	n	AUC _{0–last} (h·µg/mL)	C _{max} (µg/mL)	t _½ (h)
10 mg	20	684 ± 137	3.50 ± 0.96	398 ± 125
30 mg	20	1,894 ± 365	9.13 ± 3.29	374 ± 117
100 mg	20	7,123 ± 1,195	33.5 ± 6.5	359 ± 107

Exposure scales roughly dose-proportionally (~3× and ~10× from 10 → 30 → 100 mg). C_{max} and t_½ are similar across groups, as expected for IV bolus with BSV on clearance/volume.

(3) Exposure–Response

- Exposure: individual AUC_{0–last} from NCA
- Response: binary, target rates 10% / 50% / 90% at 10 / 30 / 100 mg (seed 42 → observed 25% / 60% / 85%)
- Best model: logistic (AIC 74.0; beat linear, E_{max}, I_{max})
- EC₅₀: 2,146 h·µg/mL (slope 0.0004)
- Plot: `/figures/ER_20260617_032936.png`

The full PK → NCA → ER pipeline ran end-to-end through PMxAgent with no manual data handling.

Figure 5: Structured case study results summary generated by the Cursor agent upon completion of the four-step workflow. The report documents outputs from each tool: (1) P

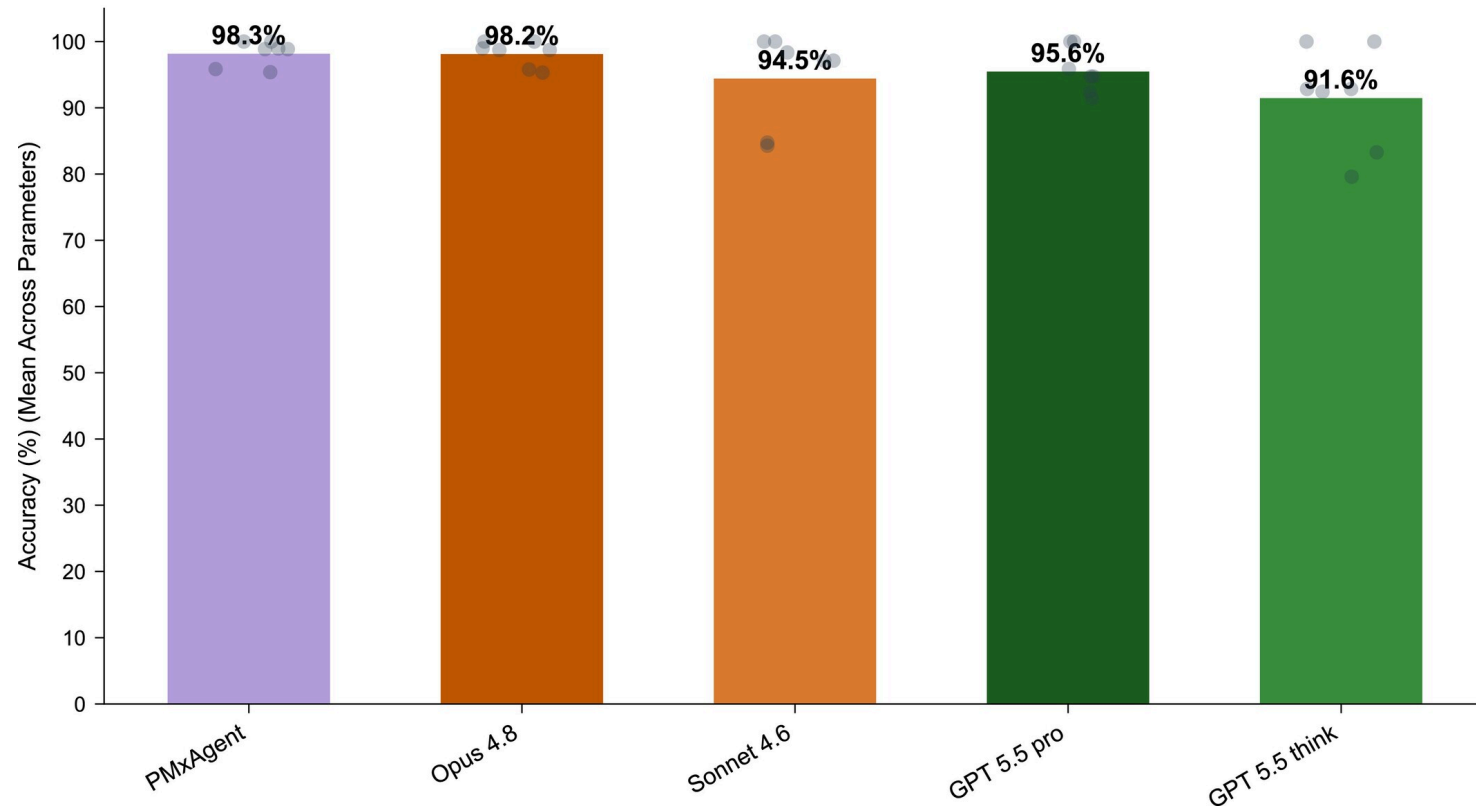


Figure 6: Overall NCA accuracy across AI agents. Bars show the mean percentage of subjects with absolute relative error ($|\text{RE}| < 1\%$) across seven primary PK parameters (Cma

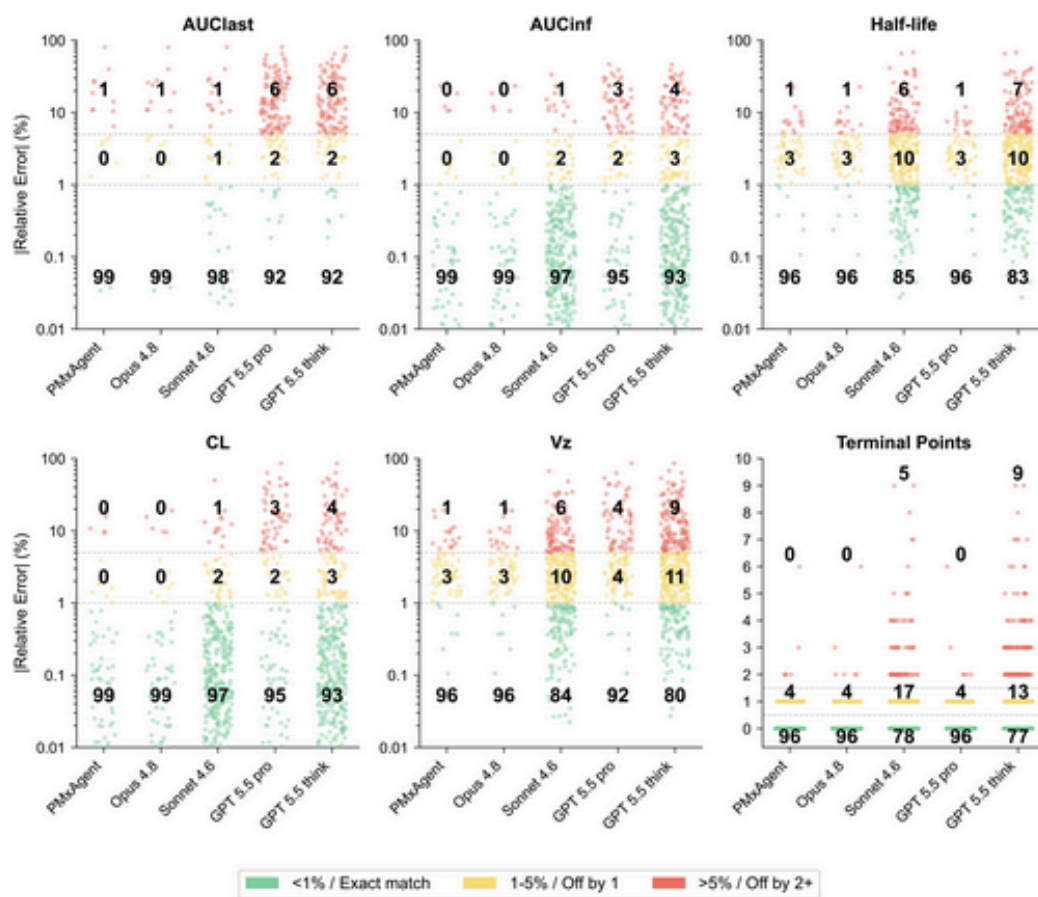


Figure 7: Per-subject error distributions by parameter and agents. Each panel shows single NCA parameters (Tmax and Cmax were excluded as they were 100% accurate across all s